Morocco NDVI Analysis

using Power BI and Google Earth Engine

Vegetation Analysis Report — 2010 to 2024

Satellite data: MODIS / VIIRS Climate data: CHIRPS, ERA5, Copernicus S5P

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I. Introduction

Vegetation dynamics represent an essential indicator for understanding the ecological state of a territory and its evolution in response to climatic factors. In Morocco, a country characterised by high climatic variability and growing pressure on natural resources, analysing vegetation dynamics is particularly strategic. The Normalized Difference Vegetation Index (NDVI) is today one of the most reliable indices for measuring the density, vigour, and health of vegetation at large scales.

This report analyses the evolution of NDVI in Morocco between **2010 and 2024**, drawing on satellite data from VIIRS and MODIS sensors, combined with climatic factors such as precipitation, luminosity, temperature, and CO_2 emissions. The objective is to identify the spatio-temporal trends of vegetation and determine which factors most significantly influence NDVI dynamics.

The study is based on a two-pronged approach:

1. A **descriptive exploration** of NDVI evolution by year, season, and region;
2. A **correlational analysis** assessing the impact of climatic parameters on vegetation cover.

II. Data Extraction (Google Earth Engine)

1. Data Sources

Data is extracted from Google Earth Engine (GEE) using Python. The following image collections are used:

Variable	GEE Collection	Aggregation
NDVI	`MODIS/061/MOD13A2`	Monthly mean
Precipitation	`UCSB-CHG/CHIRPS/DAILY`	Monthly sum (mm)
Temperature	`ECMWF/ERA5_LAND/MONTHLY`	Monthly mean (°C)
CO_2	`COPERNICUS/S5P/OFFL/L3_CO`	Monthly mean
Luminosity	`NOAA/VIIRS/DNB/MONTHLY_V1/VCMCFG`	Monthly mean

Administrative boundaries are taken from `FAO/GAUL_SIMPLIFIED_500m/2015/level1`. Values are extracted at **1 km** resolution and reduced per region using `ee.Reducer.mean()`.

2. Extraction Code

```python
import ee
import geemap.core as geemap

ee.Authenticate()
ee.Initialize(project="shining-glyph-466811-u6")
```

Listing 1: GEE initialisation

```python
COUNTRY    = 'Morocco'
START_YEAR = 2010
END_YEAR   = 2024
SCALE_M    = 1000    # ~1 km

adm1 = (ee.FeatureCollection('FAO/GAUL_SIMPLIFIED_500m/2015/level1')
          .filter(ee.Filter.eq('ADM0_NAME', COUNTRY))
          .select(['ADM0_NAME', 'ADM1_NAME']))

ic_ndvi  = ee.ImageCollection('MODIS/061/MOD13A2').select('NDVI')
ic_chirps = ee.ImageCollection('UCSB-CHG/CHIRPS/DAILY').select('precipitation')
ic_era5m  = ee.ImageCollection('ECMWF/ERA5_LAND/MONTHLY').select('temperature_2m')
```

```
ic_co     = (ee.ImageCollection('COPERNICUS/S5P/OFFL/L3_CO')
                .select('CO_column_number_density'))
ic_viirs  = ee.ImageCollection('NOAA/VIIRS/DNB/MONTHLY_V1/VCMCFG')
    .select('avg_rad')
```

Listing 2: Country, region, and dataset definitions

```
years  = ee.List.sequence(START_YEAR, END_YEAR)
months = ee.List.sequence(1, 12)

def masked_band(name):
    return ee.Image.constant(0).updateMask(ee.Image.constant(0)).
        rename(name)

def monthly_panel(y, m):
    y = ee.Number(y); m = ee.Number(m)
    start = ee.Date.fromYMD(y, m, 1)
    end   = start.advance(1, 'month')

    def safe_mean(col, name):
        return ee.Image(ee.Algorithms.If(
            col.filterDate(start, end).size().gt(0),
            col.filterDate(start, end).mean().rename(name),
            masked_band(name)))

    def safe_sum(col, name):
        return ee.Image(ee.Algorithms.If(
            col.filterDate(start, end).size().gt(0),
            col.filterDate(start, end).sum().rename(name),
            masked_band(name)))

    ndvi   = safe_mean(ic_ndvi,  'ndvi').multiply(0.0001)
    precip = safe_sum (ic_chirps, 'precip_mm')
    t2m    = safe_mean(ic_era5m, 't2m_c').subtract(273.15)
    co     = safe_mean(ic_co,    'co_col_molm2')
    viirs  = safe_mean(ic_viirs, 'viirs_rad')

    cube = ee.Image.cat([ndvi, precip, t2m, co, viirs])

    stats = (cube.reduceRegions(
                 collection=adm1,
                 reducer=ee.Reducer.mean(),
                 scale=SCALE_M,
                 tileScale=4)
             .map(lambda f: f.set({'year': y, 'month': m})))
    return stats
```

Listing 3: Monthly aggregation per region

```
ym_list = years.map(
    lambda y: months.map(
        lambda m: ee.Dictionary({'y': y, 'm': m}))).flatten()

all_stats = ee.FeatureCollection(
    ym_list.map(lambda d: monthly_panel(
        ee.Dictionary(d).get('y'),
        ee.Dictionary(d).get('m')))).flatten()

```

```
task = ee.batch.Export.table.toDrive(
    collection     = all_stats,
    description    = f"{COUNTRY}_NDVI_Rain_Temp_2010_2024",
    folder         = 'EarthEngineExports',
    fileNamePrefix = f"{COUNTRY}_NDVI_Rain_Temp_2010_2024",
    fileFormat     = 'CSV',
    selectors      = ['ADM0_NAME', 'ADM1_NAME', 'year', 'month',
                      'ndvi', 'precip_mm', 't2m_c',
                      'co_col_molm2', 'viirs_rad']
)
task.start()
```

Listing 4: CSV export to Google Drive

3. Dataset Overview

dataset_preview.png

Figure 1: Preview of `Morocco_NDVI_Rain_Temp_2010_2024.csv`.

Notes on data availability:

- **Luminosity** (VIIRS): available from **July 2011** onwards only.
- **CO_2** (Copernicus S5P): available from **2018** onwards only.

III. Data Cleaning

After loading the CSV into **Power BI**, the *Transform Data* function is used to set column types and create the following computed variables:

- **Date:** combines month and year into a single date field.
- **Season:** classifies each record into Winter, Spring, Summer, or Autumn.
- **Variation:** $\Delta\mathrm{NDVI}_K = \overline{\mathrm{NDVI}}_K - \overline{\mathrm{NDVI}}_{K-1}$
- **Variation Category:** *Increase* if $\Delta > 0$, *Decrease* if $\Delta < 0$, *None* if $\Delta = 0$.

powerbi_cleaned_data.png

Figure 2: Cleaned and enriched dataset in Power BI after column transformations.

IV. Data Visualisation

1. General NDVI Visualisation

dashboard_page1.png

Figure 3: Power BI dashboard – General NDVI Visualisation page.

a. General Overview

The national average NDVI over 2010–2024 is **0.26**, indicating overall **sparse vegetation** across Morocco. The range between the maximum (0.66) and minimum (0.08) reflects **significant heterogeneity** in vegetation cover.

b. Regional Analysis

The *"Average NDVI by Region"* map shows a clear spatial gradient: NDVI decreases moving southward and eastward. Vegetation is most abundant in the **north and west**, and sparse in the **south and east**. The top/bottom-5 region charts confirm this:

- **Highest NDVI variation (north/west):** Tanger-Tétouan, Gharb-Chrarda, Rabat-Salé-Zemmour, Grand Casablanca, Doukkala-Abda.
- **Lowest/negative variation (south/east):** Laâyoune-Boujdour, Guelmim-Es-Semara, Souss-Massa-Draâ, Oriental, Meknès-Tafilalet.

c. Seasonality Analysis

Northern and western regions display strong seasonality: NDVI peaks in **December–March** and reaches its minimum in **July**. Southern and eastern regions show almost no seasonal variation, with inter-season differences of 0.01–0.02 at most.

ndvi_season_north.png

(a) Tanger-Tétouan (north): winter–summer difference: **0.11**.

ndvi_season_south.png

(b) Laâyoune (south): winter–summer difference: **0.01**.

Figure 4: Average NDVI by season – northern vs. southern region.

The *"Average NDVI per Month"* chart confirms a national peak in **February–March** and a trough in **July–September**.

d. Annual NDVI Trend

The *"Average NDVI per Year"* time series reveals a **downward trend**:

- **Maximum:** 0.28, in 2011 and 2018.

- **Minimum:** 0.23, in 2022 and 2024 – linked to Morocco's recent drought (verified in Section IV.2).

e. Conclusion

A strong geographic divide exists in Moroccan vegetation: average NDVI of **0.36** for north/west regions vs. **0.17** for south/east regions.

kpi_north_west.png

(a) North/West: Avg 0.36 · Max 0.66 · Min 0.17

kpi_south_east.png

(b) South/East: Avg 0.18 · Max 0.44 · Min 0.08

Figure 5: NDVI summary statistics by geographic zone (Power BI region slicers).

2. Correlations with Influencing Factors

a. Precipitation

The NDVI–precipitation correlation is $\mathbf{r = 0.90}$ — an exceptionally high coefficient in environmental analysis, reflecting a very strong positive linear relationship. As precipitation increases, NDVI increases directly. No outliers are present in the scatter plot.

Conclusion

NDVI is **strongly linked to precipitation**. Precipitation is the dominant driver of vegetation density and vigour in Morocco.

b. Temperature

The NDVI–temperature correlation is $\mathbf{r = -0.36}$, a moderate negative relationship. Higher temperatures intensify water stress in semi-arid regions, reducing vegetation vigour. Regions averaging 20–22°C exhibit lower NDVI; cooler regions (14–17°C) show higher values. Temperature influences vegetation, but is not the dominant factor.

corr_precipitation.png

Figure 6: Correlation between NDVI and Precipitation by region (r = 0.90).

corr_temperature.png

Figure 7: Correlation between NDVI and Temperature by region (r = −0.36).

corr_luminosity.png

Figure 8: Correlation between NDVI and Luminosity by region (r = 0.86).

c. Luminosity

The NDVI–luminosity correlation is **r = 0.86** — a very strong positive relationship. Regions receiving greater luminosity support denser, more active vegetation. The scatter plot shows a tight, well-aligned point cloud consistent across all regions.

d. CO_2 Emissions

The raw correlation appears positive (r = 0.48). This is a **confounding effect**: Morocco's highest CO_2-emitting regions (Tanger, Casablanca, Rabat) also have the best climatic conditions for vegetation, acting as outliers that inflate the correlation. When restricted to industrial zones only, a **clearly negative relationship emerges**, confirming that CO_2 emissions negatively affect vegetation health.

corr_co2_all.png

(a) Overall correlation (r = 0.48) – confounded by industrial zones.

corr_co2_industrial.png

(b) Industrial zones only: negative correlation confirmed.

Figure 9: NDVI vs. CO_2 – full dataset vs. industrial zones only.

e. Conclusion

Conclusion

Two factors **directly and positively** drive NDVI: **Precipitation** (r = 0.90) and **Luminosity** (r = 0.86). One factor exerts a **moderate negative** effect: **Temperature** (r = −0.36). CO_2 also negatively affects NDVI once geographic confounders are controlled for.
This explains the regional patterns in Section IV.1: northern and western regions achieve higher NDVI and stronger seasonality due to more precipitation, higher luminosity, and lower temperatures. Southern and eastern regions face persistently low precipitation and high temperatures, yielding low NDVI with negligible seasonal variation.

3. General Synthesis

- **Overall trend (2010–2024):** Morocco experienced more NDVI decreases than increases. Maximum decrease: **−0.6** in 2016; maximum increase: **+0.4** in 2018.
- **2016 drought confirmed:** In 2015, total national precipitation reached its minimum (757,769 mm), directly causing the record NDVI drop in 2016 – consistent with r = 0.90.
- **2022–2024 drought verified:** Precipitation data for 2022–2024 confirms the NDVI decline hypothesis raised in Section IV.1.
- **Seasonal pattern:** Both *"Average NDVI by Season"* and *"Average Precipitation by Season"* follow the same ranking: Winter → Spring → Autumn → Summer.
- **Luminosity effect confirmed:** Years of peak luminosity consistently coincide

dashboard_synthesis.png

Figure 10: Power BI dashboard – Synthesis page.

with significant NDVI increases, validating r = 0.86.

V. Conclusion

This project provides a clear and structured picture of vegetation dynamics in Morocco from 2010 to 2024. By combining satellite-derived NDVI data with climatic variables — precipitation, temperature, luminosity, and CO_2 emissions — the analysis reveals both the spatial heterogeneity of vegetation cover and the key environmental drivers behind it.

The **north and west** of Morocco emerge as zones with the highest potential for sustained vegetation development, offering strategic opportunities for innovative agricultural projects, targeted investment, and the sustainable valorisation of natural resources. These findings provide a data-driven foundation for decision-making in environmental management and rural development policy.